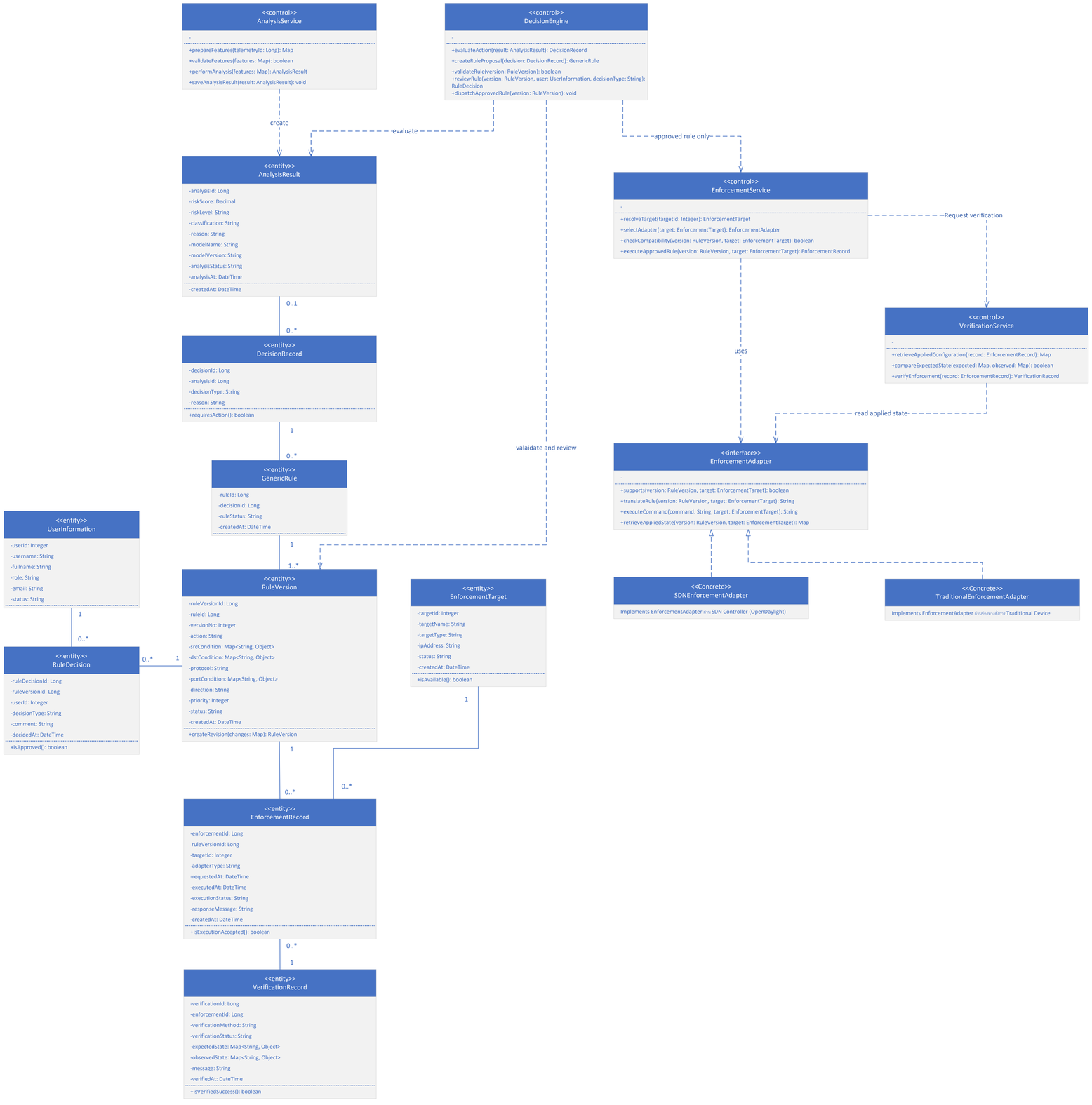

<<control>>
AnalysisService
-
+prepareFeatures(telemetryId: Long): Map
+validateFeatures(features: Map): boolean
+performAnalysis(features: Map): AnalysisResult
+saveAnalysisResult(result: AnalysisResult): void
<<control>>
DecisionEngine
-
+evaluateAction(result: AnalysisResult): DecisionRecord
+createRuleProposal(decision: DecisionRecord): GenericRule
+validateRule(version: RuleVersion): boolean
+reviewRule(version: RuleVersion, user: UserInformation, decisionType: String): RuleDecision
+dispatchApprovedRule(version: RuleVersion): void
create
evaluate
approved rule only
<<entity>>
AnalysisResult
-analysisId: Long
-riskScore: Decimal
-riskLevel: String
-classification: String
-reason: String
-modelName: String
-modelVersion: String
-analysisStatus: String
-analysisAt: DateTime
-createdAt: DateTime
0..1
0..*
<<control>>
EnforcementService
-
+resolveTarget(targetId: Integer): EnforcementTarget
+selectAdapter(target: EnforcementTarget): EnforcementAdapter
+checkCompatibility(version: RuleVersion, target: EnforcementTarget): boolean
+executeApprovedRule(version: RuleVersion, target: EnforcementTarget): EnforcementRecord
Request verification
<<control>>
VerificationService
-
+retrieveAppliedConfiguration(record: EnforcementRecord): Map
+compareExpectedState(expected: Map, observed: Map): boolean
+verifyEnforcement(record: EnforcementRecord): VerificationRecord
uses
read applied state
<<entity>>
DecisionRecord
-decisionId: Long
-analysisId: Long
-decisionType: String
-reason: String
+requiresAction(): boolean
1
0..*
valaidate and review
<<interface>>
EnforcementAdapter
-
+supports(version: RuleVersion, target: EnforcementTarget): boolean
+translateRule(version: RuleVersion, target: EnforcementTarget): String
+executeCommand(command: String, target: EnforcementTarget): String
+retrieveAppliedState(version: RuleVersion, target: EnforcementTarget): Map
<<entity>>
GenericRule
-ruleId: Long
-decisionId: Long
-ruleStatus: String
-createdAt: DateTime
<<entity>>
UserInformation
-userId: Integer
-username: String
-fullname: String
-role: String
-email: String
-status: String
1
1..*
<<entity>>
RuleVersion
-ruleVersionId: Long
-ruleId: Long
-versionNo: Integer
-action: String
-srcCondition: Map<String, Object>
-dstCondition: Map<String, Object>
-protocol: String
-portCondition: Map<String, Object>
-direction: String
-priority: Integer
-status: String
-createdAt: DateTime
+createRevision(changes: Map): RuleVersion
<<entity>>
EnforcementTarget
-targetId: Integer
-targetName: String
-targetType: String
-ipAddress: String
-status: String
-createdAt: DateTime
+isAvailable(): boolean
<<Concrete>>
SDNEnforcementAdapter
Implements EnforcementAdapter ผ่าน SDN Controller (OpenDaylight)
<<Concrete>>
TraditionalEnforcementAdapter
Implements EnforcementAdapter ผ่านช่องทางอื่นๆ Traditional Device
1
0..*
<<entity>>
RuleDecision
-ruleDecisionId: Long
-ruleVersionId: Long
-userId: Integer
-decisionType: String
-comment: String
-decidedAt: DateTime
+isApproved(): boolean
0..*
1
1
0..*
1
0..*
<<entity>>
EnforcementRecord
-enforcementId: Long
-ruleVersionId: Long
-targetId: Integer
-adapterType: String
-requestedAt: DateTime
-executedAt: DateTime
-executionStatus: String
-responseMessage: String
-createdAt: DateTime
+isExecutionAccepted(): boolean
0..*
1
<<entity>>
VerificationRecord
-verificationId: Long
-enforcementId: Long
-verificationMethod: String
-verificationStatus: String
-expectedState: Map<String, Object>
-observedState: Map<String, Object>
-message: String
-verifiedAt: DateTime
+isVerifiedSuccess(): boolean